A Prior-Odds Adjustment for the Probability of Direction in Multiple Testing

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Abstract

The probability of direction (pd) has seen rapid adoption in psychological research as an intuitive posterior summary that quantifies how strongly the posterior distribution favors one direction of an effect over the other. In practice, researchers often turn pd into a binary decision rule by applying fixed thresholds (e.g., $pd > 0.975$), declaring evidence for an effect when posterior mass concentrates strongly on one side. When this rule is applied across many parameters, it can substantially increase the risk of misleading conclusions. We argue that this problem stems from neglecting the prior probability that for every tested effect, the data-favored direction does not reflect a genuine effect. This quantity, often left implicit, becomes increasingly consequential as the number of tested effects grows. To address this, we propose a prior-odds adjustment to pd that requires researchers to explicitly specify the proportion of tested effects they expect to be null. The adjustment accounts for multiplicity without modifying the model used in parameter estimation, offers a transparent mechanism for incorporating prior skepticism about the existence of effects, and reduces the risk of false discoveries in multi-parameter settings. We evaluate its behavior through simulation across varying sample sizes and proportions of null effects, and provide an R package to facilitate applied use.

Keywords: probability of direction, multiple testing, prior-odds, FDR

Introduction

Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). This expansion has, however, been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread adoption gave way to automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). Researchers sometimes treat Bayes Factor thresholds (e.g., BF > 3) as dichotomous decision rules, or claim that an effect "exists" because the posterior probability of a directional effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). Such cutoffs are convenient, but they encourage all-or-none interpretations, and when these binary judgments are applied repeatedly across many parameters they generate a multiplicity problem no different in kind from the one long recognized in frequentist statistics.

Frequentists developed methods to address this problem, controlling either the Family-Wise Error Rate (FWER; the probability of at least one false positive) or the False Discovery Rate (FDR; the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process. When researchers examine many parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes (Altoè et al., 2020; Gelman & Carlin, 2014; Zwet & Cator, 2021). Bayesian methods can mitigate this overestimation, pulling extreme estimates toward more plausible values through informative priors (Dawid, 1994; Zwet & Gelman, 2022) and hierarchical modeling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011). Magnitude bias and false-positive accumulation are, however, distinct problems. Shrinkage offers no guarantee about the error properties of a decision rule applied across many parameters: if the same cutoff is used repeatedly (e.g., declaring an effect whenever a posterior probability exceeds 95%), the probability of spurious findings can remain substantial. Bayesian responses to this issue do exist, such as simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997).

Here we focus on an increasingly adopted decision rule based on the probability of direction (pd) (Ben-Artzi et al., 2022; Bramley & Ruggeri, 2022; Brissenden et al., 2023, 2024; Ciranka et al., 2026; Hochman et al., 2025; Makowski et al., 2020; Schurr et al., 2024; Stone-Sabali et al.,

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Code is openly available in the *multibayes* repository on GitHub at https://github.com/mar-cald/multibayes. The authors declare that they have no competing interests.

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2024). The pd quantifies the probability that an effect lies in its dominant direction (Makowski et al., 2019; Shi & Yin, 2021); researchers claim evidence for an effect when it exceeds a fixed threshold (e.g., $pd > 0.95$, $pd > 0.975$). We examine the frequentist operating characteristics[1] of such a rule, and consider how the resulting multiplicity problem can be addressed. Following the second strategy noted above, we propose a prior-odds correction for pd grounded in the global null framework of Jeffreys (1938; see also Westfall et al., 1997) and more recent Bayesian approaches to FDR control (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010; Storey, 2002, 2003). The correction requires the researcher to specify the proportion of tested effects expected to be null.

It bears emphasis that the proposed adjustment does not control error rates per se; rather, it makes explicit the prior skepticism the researcher brings to the family of hypotheses, allowing the resulting posterior probability to reflect both the observed data and the researcher's substantive beliefs about the prevalence of true effects. When those beliefs are well-calibrated to the data-generating process, error rates are naturally reduced as a consequence.

The paper proceeds as follows. We begin by introducing pd and examining its use as a test of an effect's existence, then discuss how prior choice shapes its behavior and why more skeptical within-model priors alone cannot resolve the multiplicity problem. We then develop the prior-odds adjustment, evaluate its properties through simulation, and compare its performance with two others Bayesian multiple comparison procedures. A worked example using the accompanying R package (Calderan & Gambarota, 2026) follows, after which we conclude with a general discussion.

Probability of direction

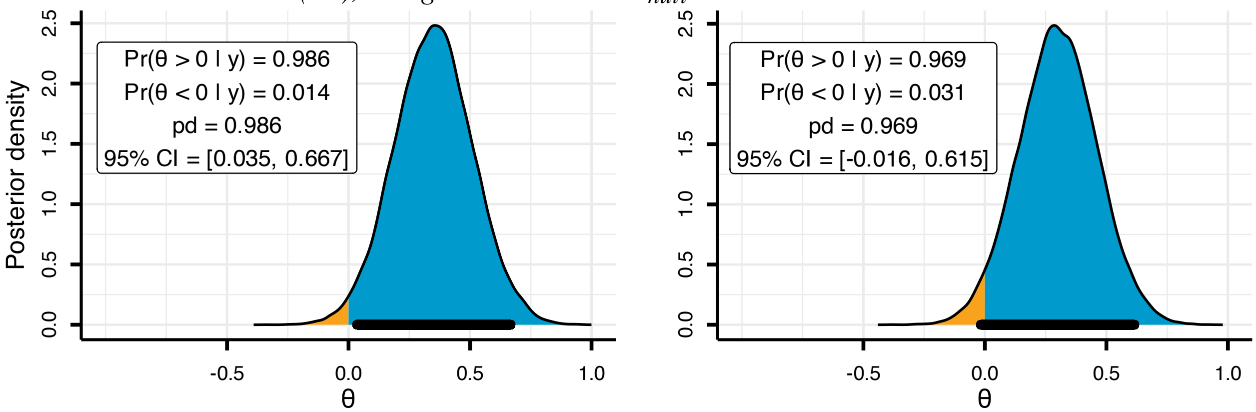
The probability of direction (pd) is a posterior summary that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities: $pd = \max(\Pr(\theta < \theta_{null} \mid y), \Pr(\theta > \theta_{null} \mid y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side). Under a uniform prior, pd is exactly the complement of the one-sided p-value for location parameters of exponential family distributions, and this relationship remains approximately valid under a broader class of priors (Casella & Berger, 1987; Marsman & Wagenmakers, 2017; Shi & Yin, 2021).

In practice, the pd is often used to summarize both the direction and existence of an effect. This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$. For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > 0 \mid y)$. An $\alpha = 0.05$ implies $c = 0.975$, therefore the rule $pd > c$ requires that $\Pr(\theta \leq 0 \mid y) < 0.025$, placing less than 2.5% of posterior mass at or below 0.

[1] Although frequentist and Bayesian frameworks rest on different foundations, examining the frequentist operating characteristics of a Bayesian procedure (how it behaves over repeated use) is a well-established practice (Müller et al., 2004; Ventz & Trippa, 2015; Zhang et al., 2019). This perspective treats long-run performance as a useful diagnostic for Bayesian inference (Gelman et al., 2013; Rubin, 1984).

Figure 1

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using c = 0.975 and $\theta_{null} = 0$.



As Figure 1 illustrates, observing $pd > 0.975$ is equivalent to excluding 0 from the central 95% equi-tailed credible interval. When the credible interval excludes the null value, one can infer that the null is relatively implausible (Box & Tiao, 1992). Given these considerations, although pd and the frequentist p-value target different quantities, it is expected that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$.

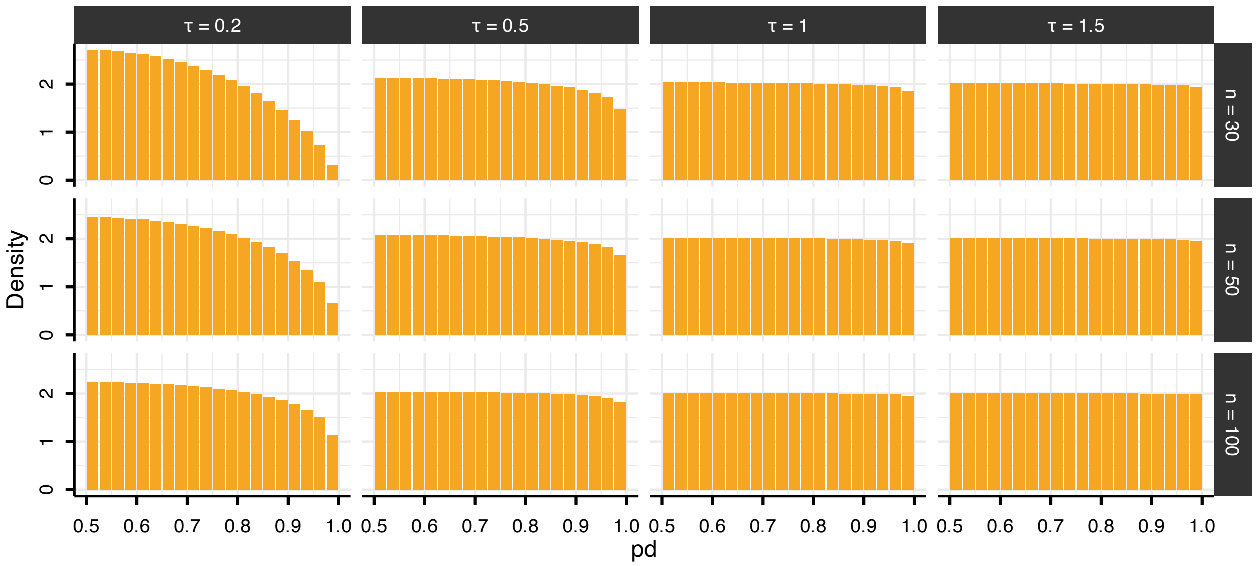
Although we express the decision threshold as $c = 1 - \alpha/2$ (with $\alpha = 0.05$) and set $\theta_{\text{null}} = 0$ throughout for ease of exposition, neither choice is restrictive. The threshold c can be set to any posterior probability cutoff, and θ_{null} can take any non-zero value, which is particularly relevant when hypotheses involve a minimum effect of practical interest (Lakens et al., 2018).

Multiplicity problem

Under the null hypothesis, p-values are uniformly distributed on $[0, 1]$. Because pd can be mapped to the p-value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$. However, because pd is a posterior probability regarding the sign of θ, its null distribution depends not only on the likelihood but also on how strongly the prior concentrates probability near 0, that is, on how strongly it "expects" near-null effects. To make this concrete, we use a conjugate normal–normal model: the observations are normal with known variance, $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$, and the effect carries a mean-zero normal prior, $\theta \sim \mathcal{N}(0, \tau^2)$. This pairing yields a normal posterior in which the prior scale governs the degree of skepticism. Varying the sample size ($n \in \{30, 50, 100\}$) and the prior scale ($\tau \in \{0.2, 0.5, 1, 1.5\}$) shows that the probability that pd exceeds 0.975 under the null is not constant (Figure 2). Under diffuse priors ($\tau \geq 1$) and/or sufficiently large samples, pd recovers the uniform null reference, with constant density on $[0.5, 1]$. A skeptical prior (e.g., $\tau = 0.2$), by contrast, induces shrinkage toward 0.5, effectively reducing the error rate.

Figure 2

Distribution of the probability of direction (pd) stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5\}$; columns). Bin width = 0.025.



While skeptical priors reduce the probability that any single test yields an extreme pd, they do not address the issue that, as the number of tests m increases, so does the probability of observing at least one extreme value among $pd_1, \ldots, pd_m$. The approach presented below addresses this at the level of the testing procedure rather than the within-model prior, by placing a separate prior on the proportion of effects expected to be null. This lets researchers who use uninformative or weakly informative priors for estimation still incorporate substantive beliefs at the family level.

As we formalize in the next section, the correction operates on the Bayes Factor comparing one directional hypothesis against the other. For pd to enter this framework, it must coincide with that Bayes Factor, and this holds only under a specific condition. Because pd is the posterior probability mass on one side of θ_{null}, it maps directly onto the posterior odds between the two directional hypotheses ($\theta > \theta_{\text{null}}$ vs. $\theta < \theta_{\text{null}}$). These posterior odds equal the Bayes Factor only when the within-model prior is symmetric around θ_{null}: symmetry makes the two directions equally plausible a priori, so the prior odds are unity and the posterior odds reduce to the ratio of marginal likelihoods. This requirement is typically satisfied by the default priors in standard Bayesian software, and it remains compatible with any degree of prior information about effect magnitude, which is expressed through θ_{null}.

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 0.5$. Without loss of generality, assume the posterior favors H_+, such that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$. The complement is then given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypotheses are therefore:

$$\frac{pd}{1-pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}$$

For a continuous posterior distribution, the probability of any single value (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero. Applying Bayes' rule, these posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})}$$

where $\mathrm{BF}_{+/-0}$ quantifies the relative likelihood of the data under H_+ versus H_{-0}. Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{1-pd}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \le 0$ (or $\theta < 0$ over $\theta \ge 0$).

Multiplicity adjustment via prior odds

For each test $i = 1, \ldots, m$, let $H_{1,i}$ denote the directional claim corresponding to the posterior-dominant sign, used to compute $pd_i = \Pr(H_{1,i} \mid y)$, with complement $\Pr(H_{0,i} \mid y) = 1 - pd_i$. If the m tests are exchangeable and $\pi_0 = \Pr(H_{0,i})$ denotes the prior probability that the data-disfavored hypothesis holds for a given test, then π_0 is the expected proportion of tests for which no genuine effect exists in the favored direction; accordingly, the expected number of such tests is $m\pi_0$.

If the tests are independent, the probability of the global null H_0, the event that no tested effect lies in its data-favored direction, is:

$$\Pr(H_0) = \pi_0^m, \qquad \text{equivalently} \qquad \pi_0 = \Pr(H_0)^{1/m}.$$

In this scenario, setting $\pi_0 = \Pr(H_{1,i}) = \Pr(H_{0,i}) = 0.5$ (the symmetric prior introduced in the previous section) assigns near-certainty to the event that at least one effect lies in its claimed direction, since $\Pr(H_0) = 0.5^m \to 0$ as m grows. A principled response is to redistribute prior probability toward the data-disfavored hypothesis for each claim as a function of family size m (Jeffreys, 1938; Westfall et al., 1997). This adjustment enters through π_0, which modifies the prior odds. For each test, the adjusted posterior odds are:

$$\frac{pd_{\mathrm{adj},i}}{1 - pd_{\mathrm{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{1 - \pi_0}{\pi_0},$$

and solving for the adjusted posterior probability gives:

$$pd_{\mathrm{adj},i} = \frac{pd_i\,(1 - \pi_0)}{pd_i\,(1 - \pi_0) + (1 - pd_i)\,\pi_0}.$$

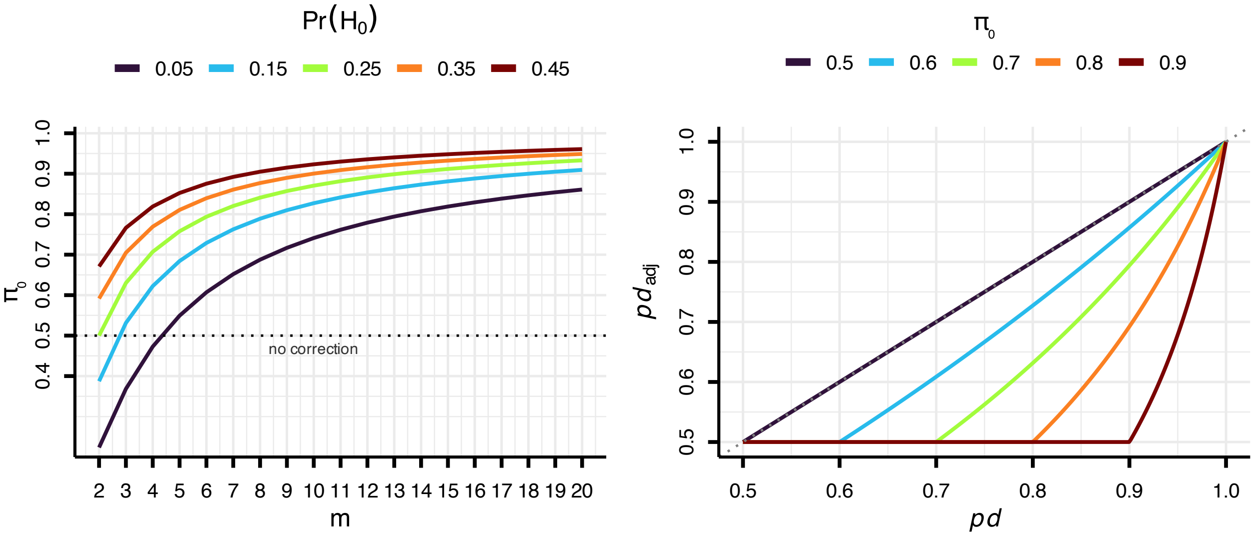
Because the prior is symmetric and pd_i is always the posterior probability of the dominant direction, the factor $(1 - \pi_0)/\pi_0$ acts as a single scalar penalty applied identically regardless of the effect's sign.

We emphasize that $H_{0,i}$ is a *directional* null, the disfavored region $\theta_i \leq \theta_{\text{null}}$ (taking the favored side as positive) covers both a truly null effect and an effect of the opposite sign that the data have mis-signed. Consequently π_0 equals the expected proportion of true (point) nulls only when such sign errors are negligible. Additionally, π_0 is a *family-level* prior, distinct from the continuous within-model prior on each θ_i, under which the point $\theta_i = \theta_{\text{null}}$ has zero mass; the procedure accordingly does not quantify evidence for the point null at any individual test, it only adjusts the weight given to directional claims in light of the expected proportion of genuine discoveries in the family.

Researchers encode their expectations through π_0 or, equivalently under independence, through the global-null prior $\Pr(H_0) = \pi_0^m$: lower values when effects are well established, higher values when uncertainty is greater, as in exploratory research. For fixed $\Pr(H_0)$, larger families imply higher π_0, then the correction acts nonlinearly on the posterior evidence, leaving larger pd_i values comparatively less affected (Figure 3).

Figure 3

The left panel shows that the implied prior probability π_0 assigned to each hypothesis $H_{0,i}$ increases with family size m, depending on the specified global null probability $\Pr(H_0)$. *The dotted line marks the boundary condition $\pi_0 = 0.5$, below which the prior would favor H_1 a priori. The right panel shows the corresponding prior-odds adjustment of pd for different π_0 values, highlighting the heterogeneous penalty: weak evidence is shrunk much more than strong evidence.*



A boundary condition arises when the specified π_0 falls below 0.5 (which, for a fixed global null, occurs when m is small and $\Pr(H_0)$ is very low), inflating the posterior odds in favor of H_1 rather than penalizing them. We therefore cap π_0 at 0.5, ensuring the adjustment acts exclusively as a conservative safeguard; when no penalty is warranted, $pd_{\text{adj},i} = pd_i$. Furthermore, since $pd \in [0.5, 1]$, the adjusted probability of direction is also bounded below by 0.5.

Error rates under the prior-odds adjustment

For a family of m such tests, let R denote the total number of rejections and V the number of false rejections (true nulls that are rejected). The quantity we want to consider is the False Discovery

Rate (FDR),

$$\mathrm{FDR} = \mathbb{E}\left[\frac{V}{\max(R,1)}\right],$$

where the $\max(R,1)$ in the denominator sets the ratio to zero when no hypothesis is rejected (Benjamini & Hochberg, 1995). The FDR can be decomposed as the product of the positive False Discovery Rate (pFDR) (Storey, 2003),

$$\mathrm{pFDR} = \mathbb{E}\left[\frac{V}{R}\,\middle|\, R > 0\right],$$

which instead conditions on at least one rejection having occurred, and the probability of making any rejection,

$$\mathrm{FDR} = \mathrm{pFDR} \times \Pr(R > 0).$$

If each hypothesis is null with prior probability π_0 and non-null with probability $1-\pi_0$, and the tests are independent and identical, the pFDR can be written as

$$\mathrm{pFDR} = \frac{\pi_0\,\alpha}{\pi_0\,\alpha + (1-\pi_0)(1-\beta)},$$

where α and $1-\beta$ are the per-test Type I error rate and power, respectively (Berger, 2023; Storey, 2003). Here π_0 is the prior probability that a hypothesis is null, whereas α and β describe the operating characteristics of the test. When the tests are heterogeneous, there is no single common (α,β) pair; one may then summarize performance by the family-level averages $\bar{\alpha}$ and $\bar{\beta}$, which yield the ratio $\mathbb{E}[V]/\mathbb{E}[R]$.

In the proposed correction the researcher's prior fixes π_0, while the same adjustment implicitly reduces α (i.e., increases c, since $c = 1-\alpha/2$). Indeed, if we write the decision rule $pd_{\mathrm{adj},i} > c$ in odds form, $\frac{pd_{\mathrm{adj},i}}{1-pd_{\mathrm{adj},i}} > \frac{c}{1-c}$, and substitute the adjusted odds defined above, the rule reduces to a condition on the unadjusted odds,

$$\frac{pd_i}{1-pd_i} > \frac{c}{1-c}\cdot\frac{\pi_0}{1-\pi_0},$$

which, returned to the probability scale, is a stricter threshold on the raw pd_i,

$$pd_i > c^*, \qquad c^* = \frac{c\,\pi_0}{(1-c)(1-\pi_0) + c\,\pi_0}.$$

A higher threshold lowers the per-test probability of rejection under the null; this per-test rate equals $\alpha = 2\bigl(1-c^*\bigr)$ when the within-model prior is diffuse or n is large (so that the null distribution of pd is approximately uniform on $[0.5, 1]$), and is smaller under an informative prior, which shrinks the posterior toward the null. The FDR, by contrast, cannot be reduced to a single number, as it depends on the power of the individual tests, so we make no claim about its absolute level. In summary, the adjustment is equivalent to a stricter per-test threshold $c_* > c$, so for $\pi_0 > 0.5$ it lowers the per-test false-positive rate and the expected number of false discoveries relative to the unadjusted rule[2].

[2]The adjustment targets the FDR, not the FWER (the probability of at least one false positive). A test produces a

Simulation Study

We conducted a fully crossed simulation study, with 10,000 replications per cell, to examine the adjustment under both correctly specified and misspecified priors. We varied the per-test null prior supplied to the procedure, $\pi_0 \in \{0.50, 0.6, \ldots, 0.9\}$; the true probability that each hypothesis is null, $\pi_{0\text{true}} \in \{0.5, 0.6, \ldots, 1.0\}$; the family size $m \in \{4, 8, 10, 20\}$; and the sample size $n \in \{30, 50, 100\}$. At each replication the number of true null effects ($\theta_i = 0$) was drawn from Binomial$(m, \pi_{0\text{true}})$ and the remaining effects set to $\theta_i = 0.3$; observations were generated as $y_i \sim \mathcal{N}(\theta_i, 1)$, and the posterior obtained from a conjugate normal–normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, 2)$. We declared support for the alternative when $pd > 0.975$ and evaluated the FDR, the FWER, and average power, the proportion of true effects detected (Benjamini & Hochberg, 1995; Izmirlian et al., 2025). The diagonal $\pi_0 = \pi_{0\text{true}}$ corresponds to a correctly specified prior.

Figure 4 shows the FDR of the procedure across the grid. The column $\pi_0 = 0.5$ is the unadjusted rule: there the prior odds are one, no correction is applied ($pd_{\text{adj}} = pd$), and the FDR inflates with the family size m and the true null proportion $\pi_{0\text{true}}$, reaching 0.64 when the family is large and most effects are null. Increasing π_0 brings the FDR under control: it is held low whenever the specified π_0 is at least the true proportion, and inflates only under substantial underestimation, when a small π_0 is paired with a large $\pi_{0\text{true}}$ at large m. This control trades off power, which decreases as π_0 increases; because average power does not vary with m (for fixed $\pi_{0\text{true}}$ and π_0), we display $m = 20$ as a representative condition (Figure 4).

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false positive only when it is both null and rejected, with probability $\pi_0\,\alpha^*$, so across m independent tests FWER $= 1-(1-\pi_0\,\alpha^*)^m$. Controlling it at a target α gives $\alpha^* = \frac{1-(1-\alpha)^{1/m}}{\pi_0}$. Because the two-sided rule $pd > c$ has per-test Type I rate $\alpha^* = 2(1-c)$ (the convention $c = 1-\alpha/2$ used throughout), inverting this relation turns the family-wise correction into a single cutoff on the raw pd, $c_{\text{fwer}} = 1 - \alpha^*/2 = 1 - \frac{1-(1-\alpha)^{1/m}}{2\,\pi_0}$. This uses the same identity $\alpha^* = 2(1-c)$, so it likewise assumes a diffuse prior or large n and independent tests; an informative prior lowers the per-test rate and makes c_{fwer} conservative, controlling the FWER at or below the target.

Figure 4

False Discovery Rate (FDR, Panel A) and average power (Panel B) of the adjusted pd procedure as a function of the specified per-test null prior π_0 (x-axis) and the true null probability $\pi_{0_{true}}$ (y-axis), by family size m (rows, for Panel A) and sample size n (columns). The column $\pi_0 = 0.5$ is the unadjusted rule, where the prior odds are one and no correction is applied ($pd_{adj} = pd$); the diagonal $\pi_0 = \pi_{0_{true}}$ is a correctly specified prior. Each cell is the empirical FDR/power over 10,000 replications.

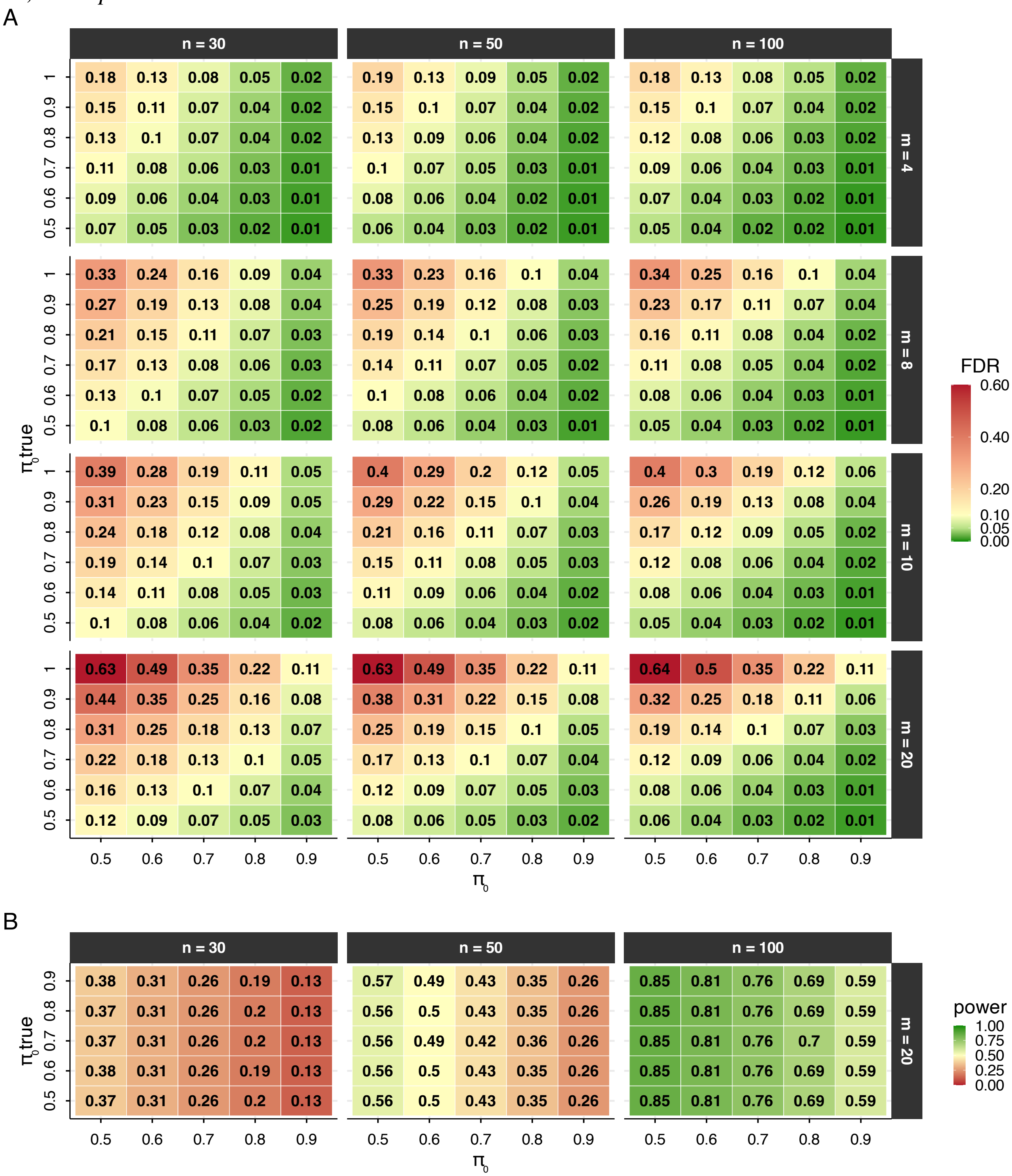


Figure 5 shows the FWER of the same rule (for $n = 50$, as a representative case). Unlike the FDR, the FWER is not held low by the per-test adjustment: it grows with the family size m, reaching 0.64 for the unadjusted rule with a large family of mostly null effects. Increasing π_0 lowers it substantially, at $\pi_0 = 0.9$ it stays at or below about 0.10.

Figure 5

Family-wise error rate (FWER) of the adjusted pd procedure as a function of the specified per-test null prior π_0 (x-axis) and the true null probability $\pi_{0_{true}}$ (y-axis), by family size m (columns). The column $\pi_0 = 0.5$ is the unadjusted rule. Each cell is the empirical FWER over 10,000 replications. Because the per-test adjustment targets the FDR, the FWER is reduced by larger π_0 but still grows with the family size m.

Comparison with other Bayesian methods

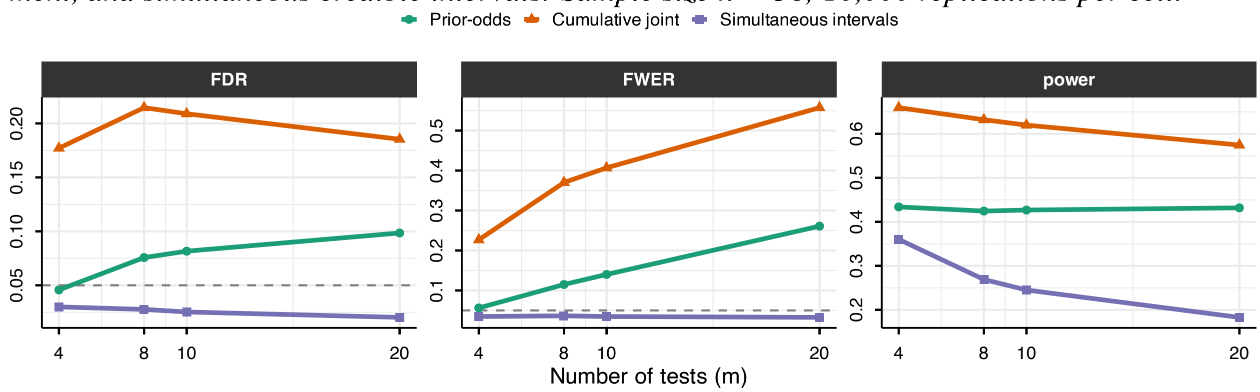
Two further Bayesian summaries have been proposed to address multiplicity. The first constructs a cumulative joint posterior probability: one accepts the strongest k claims whose joint posterior probability is at least $1 - \alpha$ (Gelman & Tuerlinckx, 2000; Pruzek, 2016). The second relies on simultaneous credible intervals, declaring an effect whenever its $1 - \alpha$ simultaneous band excludes the null value (Box & Tiao, 1992; Schnell et al., 2016; Schnell et al., 2020). To benchmark all three approaches, we replicate the simulation described above with $n = 50$ and $\pi_{0\text{true}} = 0.7$ held fixed, and vary the family size m; the prior-odds adjustment is evaluated at $\pi_0 = \pi_{0\text{true}}$, representing best-case calibration. Figure Figure 6 displays the resulting FDR, FWER, and power for each rule across values of m.

The simultaneous intervals are the strictest, controlling the FWER (and hence the FDR) below the nominal level throughout (as $\pi_{0\text{true}} = 0.7$, and not 1). The cumulative joint statement is the least restrictive, with the FDR reaching 0.21, and FWER 0.55. This is expected, because the cumulative joint is a sign-error (Type S) statement, thus it controls the family-wise probability of a wrong direction, which is less restrictive compare to Type I error control.

These choices trade off against power (rightmost panel): the cumulative joint detects the most true effects (around 0.63), the prior-odds procedure's power is essentially constant in m (around 0.42), and the strict simultaneous intervals lose power as the family grows (from 0.36 at $m = 2$ to 0.18 at $m = 20$). The choice among them is thus a deliberate trade between power and the error one most wants to limit.

Figure 6

False Discovery Rate (FDR), Family-Wise Error Rate (FWER) and average power of three Bayesian rules as a function of the number of tests m: the prior-odds procedure, the cumulative joint statement, and simultaneous credible intervals. Sample size n = 50; 10,000 replications per cell.



Application: worked example with the R Package `multibayes`

The procedure is implemented in the `multibayes` R package, which provides functions for computing adjusted pd values from either raw posterior draws or a vector of unadjusted pd values. The core function, `pd.adjust()`, accepts posterior draws directly and computes pd values internally when needed. The prior is supplied in either of two ways: directly as the per-test null prior π_0, or as the global-null prior $\Pr(H_0)$, from which $\pi_0 = \Pr(H_0)^{1/m}$ is recovered under independence. The output includes the adjusted pd values, the tested null value, the per-test null probability π_0, and the number of tests m. Unadjusted pd values are retained alongside adjusted ones to facilitate direct comparison; when posterior draws are provided as input, the posterior median for each parameter is also returned. `pd.adjust()` returns the per-test correction, and `order = TRUE` sorts the hypotheses by strength of evidence. Family-wise summaries across hypotheses—the cumulative joint statement and simultaneous credible intervals—are provided separately by the companion function `joint()`. The decision threshold may be given either as a pd cutoff c (argument `threshold`, default 0.975) or, equivalently, as a nominal per-test rate (argument `alpha`, with $c = 1 - \alpha/2$); in either case the function reports the implied adjusted cutoff c^* on the raw pd, which summarizes what the chosen π_0 implies.

For this example, we use the openly available dataset from Lin et al. (2020), who investigated the effect of ego depletion on conflict resolution across four studies. In their paradigm, an initial high-demand (vs. low-demand) task was followed by a Stroop task. For simplicity, we focus on Study 1, which tested the effects of congruency and ego depletion (within subjects). The dataset is readily accessible via the `afex` package (Singmann et al., 2016) and was chosen for this reason; this example is not intended as a critique of the original work. This dataset includes data from 253 participants; for this example, we randomly selected 80 participants.

```
# install.packages('remotes')
remotes::install_github("mar-cald/multibayes")
library(multibayes)
library(brms)
```

```r
library(dplyr)

data_stroop <- subset(afex::stroop, subset = study == 1 & pno %in% sample(levels(pno),
    size = 80)) |>
    # remove useless levels id
dplyr::mutate(id = droplevels(pno)) |>
    # select relevant variables
dplyr::select(id, condition, congruency, trialnum, acc, rt) |>
    # previous cong
group_by(id, condition) |>
    mutate(prev_cong = lag(congruency), trial = scale(trialnum)[, 1]) |>
    ungroup() |>
    na.omit()
```

Given the repeated-measures structure of the data and the large number of trials per participant, we fit a multilevel generalized linear model using `brms` (Bürkner, 2017) on reaction time (RT) data. A Lognormal model was specified, with default flat priors; these were adopted not as a principled recommendation, but to illustrate the method's behavior under the most common applied scenario, which also represents the condition where the prior-odds adjustment is most suggested.

The model includes the effect of congruency (congruent vs. incongruent), previous congruency (congruent vs. incongruent), condition (depletion vs. control), and trial number (scaled within participant and condition). The sequential congruency effect was modeled as the interaction between congruency and previous congruency; the hypothesis that depletion amplifies the congruency effect over time was captured by the three-way congruency × condition × trial interaction. This yielded nine hypotheses (Table 1).

Although the theoretical hypotheses are directional (e.g., incongruent trials are expected to slow responses), pd is computed uniformly as $\max\big(\Pr(\hat{\theta} > \theta_{\text{null}}),\ \Pr(\hat{\theta} < \theta_{\text{null}})\big)$ for all hypotheses.

Table 1

Tested hypotheses for RT, evaluated on the log scale of the posterior.

Hypothesis	$\boldsymbol{\theta_{\text{null}}}$
H_1: congruency effect	0
H_2: previous congruency effect	0
H_3: condition effect	0
H_4: trial number effect	0
H_5: congruency × previous congruency interaction	0
H_6: congruency × condition interaction	0
H_7: congruency × trial interaction	0
H_8: trial × condition interaction	0
H_9: congruency × condition × trial interaction	0

We retained only correct trials preceded by a correct trial (i.e., error and post-error trials were excluded), yielding $5,965$ observations.

```
data_rt <- data_stroop |>
    group_by(id) |>
    filter(acc == 1 & lag(acc) == 1) |>
    ungroup()

# set contrast
contrasts(data_rt$congruency) <- contr.sum(2)/2
contrasts(data_rt$prev_cong) <- contr.sum(2)/2
contrasts(data_rt$condition) <- contr.sum(2)/2

mod_rt <- brm(formula = rt ~ 1 + congruency * prev_cong + congruency * trial * condition
    (1 + congruency + prev_cong + trial + condition || id), family = lognormal(),
    data = data_rt, cores = 3, chains = 3, iter = 8000)
```

Since `pd.adjust()` computes *pd* values automatically from a matrix of posterior draws, we extracted the relevant draws from the model.

```
draws_rt <- as.data.frame(mod_rt)
par_names <- colnames(draws_rt)[2:10]
draws_rt <- draws_rt[, par_names]
pd.adjust(draws = draws_rt, pi0 = 0.7, null.value = 0)
```

```
Prior-odds adjusted probability of direction
Tests: 9  |  pi0 = 0.7
Decision: raw pd > 0.9891 (adjusted from 0.975); effective per-test alpha
0.0217 (from 0.05)

                                median.est null.value     pd pd.adj direction
b_congruency1                      -0.1439          0 1.0000 1.0000 two.sided
b_prev_cong1                       -0.0166          0 0.9803 0.9553 two.sided
b_trial                             0.0119          0 0.9369 0.8642 two.sided
b_condition1                        0.0063          0 0.6509 0.5000 two.sided
b_congruency1:prev_cong1            0.0223          0 0.9583 0.9079 two.sided
b_congruency1:trial                -0.0033          0 0.7206 0.5250 two.sided
b_congruency1:condition1           -0.0166          0 0.9291 0.8488 two.sided
b_trial:condition1                  0.0071          0 0.8936 0.7825 two.sided
b_congruency1:trial:condition1     -0.0028          0 0.5961 0.5000 two.sided
```

The function was called with a per-test null prior of $\pi_0 = 0.7$ (shown in the printed header), encoding the skeptical expectation that roughly 70% of the tested effects are null. The header also reports the equivalent decision this implies: at the default cutoff $c = 0.975$, a raw *pd* must now exceed the stricter adjusted threshold $c^* = 0.989$.

The adjustment's effect varies with the strength of the underlying evidence. The predictor `b_congruency1` carries $pd = 1$ and is left untouched ($pd_{\text{adj}} = 1$). Terms with lower but still substantial evidence are pulled down only mildly: `b_prev_cong1` (0.980 → 0.955), `b_trial` (0.937 → 0.864), `b_congruency1:prev_cong1` (0.958 → 0.908), `b_congruency1:condition1` (0.929 → 0.849), and `b_trial:condition1` (0.894 → 0.783). In contrast, parameters with weak evidence are shrunk substantially toward 0.5: `b_condition1` (0.651 → 0.500), `b_congruency1:trial` (0.721 → 0.525), and the three-way interaction `b_congruency1:trial:condition1` (0.596 → 0.500).

The "family-wise" picture can be obtained separately with `joint()`, default compute joint cumulative probabilities:

```r
joint(draws_rt, null.value = 0)
```

```
                               median.est null.value     pd direction joint_cum
b_congruency1                     -0.1439          0 1.0000 two.sided    1.0000
b_prev_cong1                      -0.0166          0 0.9803 two.sided    0.9803
b_congruency1:prev_cong1           0.0223          0 0.9583 two.sided    0.9412
b_trial                            0.0119          0 0.9369 two.sided    0.8812
b_congruency1:condition1          -0.0166          0 0.9291 two.sided    0.8191
b_trial:condition1                 0.0071          0 0.8936 two.sided    0.7340
b_congruency1:trial               -0.0033          0 0.7206 two.sided    0.5366
b_condition1                       0.0063          0 0.6509 two.sided    0.3532
b_congruency1:trial:condition1    -0.0028          0 0.5961 two.sided    0.2232
```

With the rows sorted by decreasing pd, the `joint_cum` column reports the cumulative joint posterior probability that the strongest k directional claims hold simultaneously, the posterior probability that every claimed direction is correct at once. A high value therefore licenses accepting that whole set of claims together. Here the two strongest hold jointly with probability 0.98 and the three strongest with 0.94, after which the joint confidence lowers as weaker claims are added, indicating how many of the leading effects can be reported as a jointly supported set.

For simultanoues equi-tailed intervals `joint(interval = TRUE)`

```r
joint(draws_rt, interval = TRUE)
```

```
                                     lower          est        upper prob
b_congruency1                  -0.191547431 -0.143902841 -0.097013276 0.95
b_prev_cong1                   -0.040931203 -0.016649556  0.006550712 0.95
b_trial                        -0.010103453  0.011930699  0.034360234 0.95
b_condition1                   -0.041911018  0.006263499  0.055382483 0.95
b_congruency1:prev_cong1       -0.012481840  0.022342004  0.056749925 0.95
b_congruency1:trial            -0.019082492 -0.003316115  0.012417422 0.95
b_congruency1:condition1       -0.048854131 -0.016551526  0.015382114 0.95
b_trial:condition1             -0.008945797  0.007125574  0.023144055 0.95
b_congruency1:trial:condition1 -0.034805697 -0.002829134  0.028034847 0.95
```

Discussion

The probability of direction (pd) is the posterior mass on the data-favored side of a reference null value, equivalently, the larger of the two one-sided posterior probabilities. Under within-model prior symmetry each such data-selected hypothesis is evaluated under prior odds of one, so that the procedure attributes equal a priori plausibility to the direction selected after inspection of the data. In this scenario, if multiple hypothesis are tested, the procedure becomes a priori almost certain that at least one favored direction is genuine. The approach developed here addresses this by recasting multiplicity as a problem of prior specification: building on the global-null formulation of Jeffreys (1938; see also Westfall et al., 1997) and Bayesian approaches to FDR control (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010; Stephens, 2017; Storey, 2002, 2003), the correction requires researchers to specify the prior proportion of effects expected to be null. Making this prior explicit serves two purposes. For researchers using diffuse within-model priors, it provides a principled way to encode skepticism at the family level without committing to informative priors on individual parameters. For researchers who judge the implicit prior appropriate and decline the adjustment, it nonetheless renders that commitment transparent.

Related corrections have been implemented for specific testing contexts, such as post-hoc tests in JASP (Jong, 2019; van den Bergh et al., 2020). The present procedure extends this logic to pd and provides a general-purpose implementation for applied use, applicable to directly to posterior draws or any posterior-derived pd, including cases where pd values arise from multiple separate analyses within the same manuscript (Blondé et al., 2021; Ciranka et al., 2026). This raises the question of how to define the hypothesis family to which the correction applies (Hochberg & Tamhane, 1987; Shaffer, 1995). Although no universal consensus exists regarding what constitutes a family, we favor the criteria proposed by Westfall et al. (2011), where a family is a set of questions that *"form a natural and coherent unit, [are] considered simultaneously in the decision-making process, and/or present a set over which one plays pick-the-winner"*. Regardless of the chosen definition, authors should clearly specify their chosen family and explain how it serves the study's substantive goals.

We further recommend that researchers justify their choice of π_0, the prior proportion of effects expected to be null, by drawing on theory, expert elicitation, or domain expertise (O'Hagan, 2019). More generally, a high prior probability suits exploratory settings with many uncertain hypotheses, while a lower value is warranted in confirmatory settings. For applied reporting, the unadjusted pd should accompany one multiplicity-adjusted measure, with the number of tests and π_0 clearly stated.

Although our approach derives from the Bayesian FDR literature, it differs from it in a fundamental respect. There, π_0 itself carries a prior distribution and is typically estimated from the data; here, it is specified by the user. The good frequentist properties of our procedure therefore hold only when the user-specified π_0 is approximately correct. A researcher may instead deliberately adopt a more skeptical prior, tightening control of Type I error at the cost of higher Type II error; the optimal value depends on the relative costs of the two.

Two limitations of the present framework warrant attention. First, the procedure assumes that all hypotheses are exchangeable. Exchangeability means each hypothesis receives the same prior penalty, even when some effects are regarded as more plausible a priori than others. Because the penalty shrinks weak evidence far more than strong evidence, genuine effects with strong support still survive; the price of a uniform proportion is therefore a modest loss of power on hypotheses one believes more likely to be true, rather than a loss of error control, which is preserved as long as the specified proportion is reasonable. Accommodating differential prior plausibility would require specifying and calibrating a separate π_0 for each test, with the realized error rate then governed by every individual value rather than by the family average.

Second, it does not provide evidence for a point null hypothesis. The procedure evaluates directional claims defined by inequalities (e.g., $\theta > \theta_{\text{null}}$ vs. $\theta \leq \theta_{\text{null}}$), and while the complementary hypothesis includes the null value, that value receives no special status; non-rejection therefore reflects the absence of a directional discovery, not evidence in favor of $\theta = \theta_{\text{null}}$. It bears noting, however, that the prior-odds correction introduced here is not novel (Westfall et al., 1997): it reflects a foundational principle of hypothesis testing — namely, that prior probabilities should be combined with all available evidence (Dienes, 2016) — and can be extended to other Bayes factor frameworks (Heck et al., 2023; Ly et al., 2016; Morey & Rouder, 2011). When joint modeling is feasible, a principled alternative is to integrate evidence across hypotheses through joint posterior probabilities (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020).

As Bayesian methods become increasingly common in psychological research, often in pragmatic workflows relying on default priors and fixed decision thresholds, the implications of multiplicity should be explicitly recognized. Additionally, although priors may be well-calibrated at the level of individual parameters, they do not by themselves account for multiplicity when many parameters are considered simultaneously. Ignoring this aspect leads fixed pd thresholds to systematically overstate evidence for directional effects, especially as the number of tests grows. The adjustment proposed here addresses this gap by incorporating prior beliefs about the prevalence of null effects directly into pd-based decisions. By doing so, it provides a principled and transparent way to account for multiplicity without altering the estimation model, yielding more reliable inferences in multi-parameter settings and reducing the risk that apparent effects reflect chance rather than substantive signals.

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